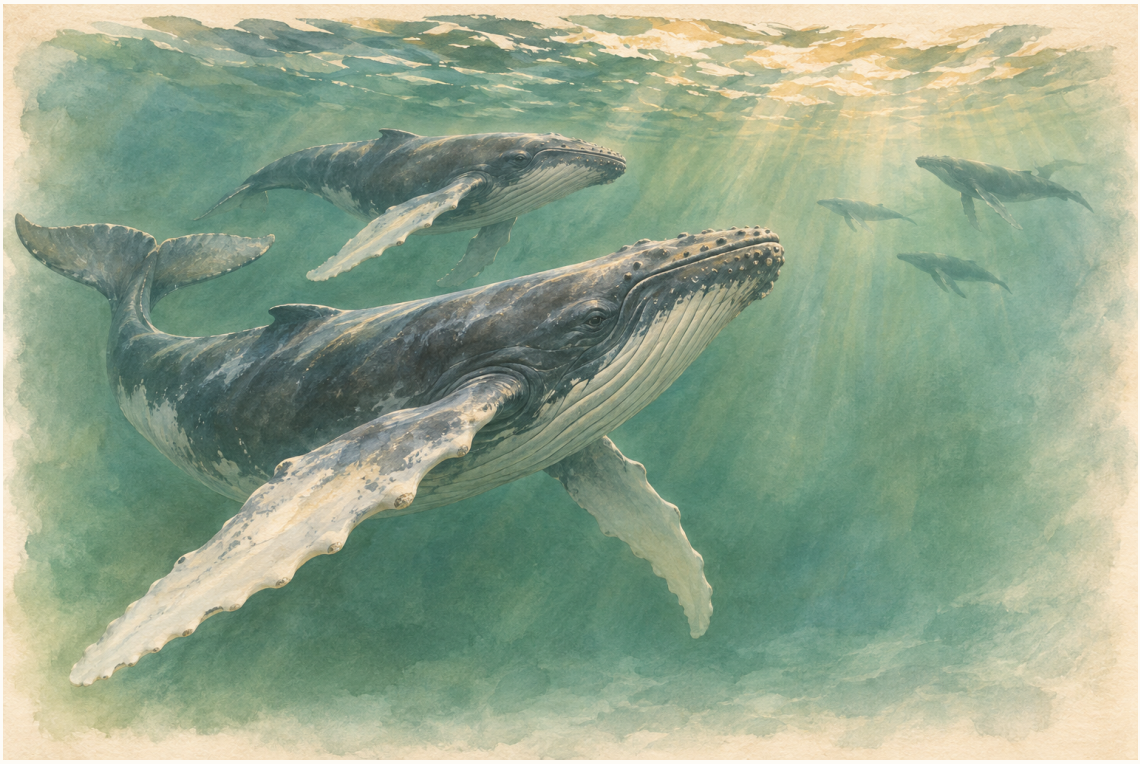
目录

第7册

61	鲸歌易帜 *A Change of Anthem in the Deep*	2
62	为机器点燃思想的人 *The Man Who Gave Machines a Mind*	9
63	没有到来的零日 *The Day Zero That Never Came*	16
64	改变人类命运的霉菌 *The Mould That Changed the World*	23
65	一封信惊醒了达尔文 *The Letter That Startled Darwin*	30
66	入侵澳洲的毒蟾 *The Toad That Invaded Australia*	36
67	从一道划痕到天花绝迹 *From a Scratch to a World Without Smallpox*	43
68	阿波罗13号：成功的失败 *Apollo 13: The Successful Failure*	49
69	拯救天空 *Saving the Sky*	56
70	写给外星人的一张唱片 *The Record Humanity Threw into the Stars*	63

鲸歌易帜

A Change of Anthem in the Deep



新歌如何跨过海洋，传遍鲸群？

英文原文

Each winter, **male humpback whales**, and only males, fill their **breeding grounds** with a **haunting melody**. **Sound units constitute** phrases, and phrases build **themes** in an **elaborate hierarchy**. Each **rendition** lasts minutes, and the song can **recur** for hours, probably for **courtship**. All the males in a **population** sing an **identical** version, which **evolves subtly** each year.

That is the rule. **Off** eastern Australia, it was broken. From 1995 to 1998, **researchers** used hydrophones, or underwater **microphones**, to record humpbacks on their **seasonal migration** along the **coastline**. In the mid-1990s, **merely** two of 82 singers **emitted** a **distinctive** new song, **utterly unlike** the **reigning chorus**.

Within about two years, **virtually** every singer had **converted**, and by 1998 the old song had been **displaced wholesale**. A study published in Nature in November 2000 found that the new song **resembled** that of the Indian Ocean population off western Australia, apparently **ferried** east by **a handful of** visiting **performers**.

Nor was it a **freak**. In April 2011, a study in Current Biology **analysed** over a decade of **recordings** from six South **Pacific** populations. Song types **primarily** arose off eastern Australia, then **rippled eastward** through **successive** populations to French **Polynesia**, **generally** crossing the **region** within about two years.

Such **rapid change**, too fast for genetic **evolution**, points to **imitation** rather than **inheritance**, a **persuasive** case of **cultural transmission**. Why males sing, and why they **overhaul** their songs, remain **controversial**. One idea is that **novelty** itself **spurs** change. Some **scholars** are **reluctant** to **define** this **as** culture in the human sense. Whatever the label, a few **outsiders** off eastern Australia **reshaped** an entire population's music.

中文译文

每年冬天，雄性座头鲸，而且只有雄性，会让它们的繁殖地回荡着一段萦绕于心头的旋律。若干声音单元构成乐句，乐句又组成主题，形成一个复杂精细的层级结构。每一次表演/演奏持续数分钟，而这首歌能反复出现数个小时，很可能是为了求偶。同一种群中的所有雄性都唱着完全相同的版本，而它每年都不易察觉地演变。

这就是规则。在澳大利亚东部海域，这个规则被打破了。1995年至1998年间，研究人员用水听器，即水下麦克风，录下了沿海岸线进行季节性迁徙的座头鲸。20世纪90年代中期，82只歌者中仅有两只发出了一首与众不同的新歌，完全不像占据主导地位的合唱。

大约两年之内，几乎每只歌者都已转变；到1998年，旧歌已被全盘取代。2000年11月发表于《自然》（Nature）的一项研究发现，这首新歌与澳大利亚西部海域印度洋种群的歌相似，似乎是由一小批来访的歌手向东带来的。

这也并非不寻常的事。2011年4月，《当代生物学》（Current Biology）上的一项研究分析了南太平洋六个种群十余年的录音。歌型主要起源于澳大利亚东部海域附近，然后向东如同涟漪般扩散，穿过相继的一个个种群传至法属波利尼西亚，通常在大约两年内横跨该区域。

如此快速的变化，对基因演化来说太快了，指向的是模仿而非遗传，即文化传递的一个有说服力的例证。雄性为何歌唱，以及它们为何要全面改造它们的歌曲，仍然存在争议。一种观点是，新奇本身推动了变化。一些学者不愿把这种现象界定为人类意义上的文化。不管叫什么名号，澳大利亚东部海域的一小批外来者，重塑了整个种群的音乐。

词汇与表达

male adj.
雄性

humpback whales n.
座头鲸

breeding grounds n.
繁殖地

haunting adj.
萦绕于心头的

melody n.
旋律

Sound units n.
声音单元

constitute v.
构成
Sound units constitute phrases,

themes n.
主题

elaborate adj.
复杂精细的
themes in an elaborate hierarchy.

hierarchy n.
层级结构
themes in an elaborate hierarchy.

rendition n.
表演/演奏

recur v.
反复出现
and the song can recur for hours,

courtship n.
求偶
for hours, probably for courtship.

population n.
种群

identical adj.
完全相同的
population sing an identical version,

evolves v.
演变

subtly adv.
不易察觉地

Off prep.
在……海域

researchers n.
研究人员

microphones n.
麦克风

seasonal adj.
季节性

词汇与表达（续）

migration n.
迁徙
on their seasonal migration along the

coastline n.
海岸线

merely adv.
仅

emitted v.
发出了

distinctive adj.
与众不同的
emitted a distinctive new song,

utterly unlike adv.
完全不像

reigning adj.
占据主导地位的

chorus n.
合唱

virtually adv.
几乎
two years, virtually every singer had

converted v.
转变

displaced v.
取代
old song had been displaced wholesale.

wholesale adv.
全盘

a handful of n.
一小批
by a handful of visiting performers.

resembled v.
与……相似
the new song resembled that of the

ferried v.
带来

performers n.
歌手

freak n.
不寻常的事

analysed v.
分析了

recordings n.
录音

Pacific n.
太平洋

primarily adv.
主要
Song types primarily arose off eastern

rippled v.
如同涟漪般扩散

词汇与表达（续）

eastward adv.
向东

successive adj.
相继的
through successive populations to

Polynesia n.
波利尼西亚

generally adv.
通常

region n.
区域

rapid change phr.
快速的变化

evolution n.
演化
too fast for genetic evolution,

imitation n.
模仿
to imitation rather than inheritance,

inheritance n.
遗传
to imitation rather than inheritance,

persuasive adj.
有说服力的

cultural adj.
文化

transmission n.
传递
case of cultural transmission.

overhaul v.
全面改造
why they overhaul their songs, remain

controversial adj.
存在争议
their songs, remain controversial.

novelty n.
新奇
is that novelty itself spurs change.

spurs v.
推动了

scholars n.
学者

reluctant adj.
不愿
Some scholars are reluctant to define

define as v.
界定为

outsiders n.
外来者

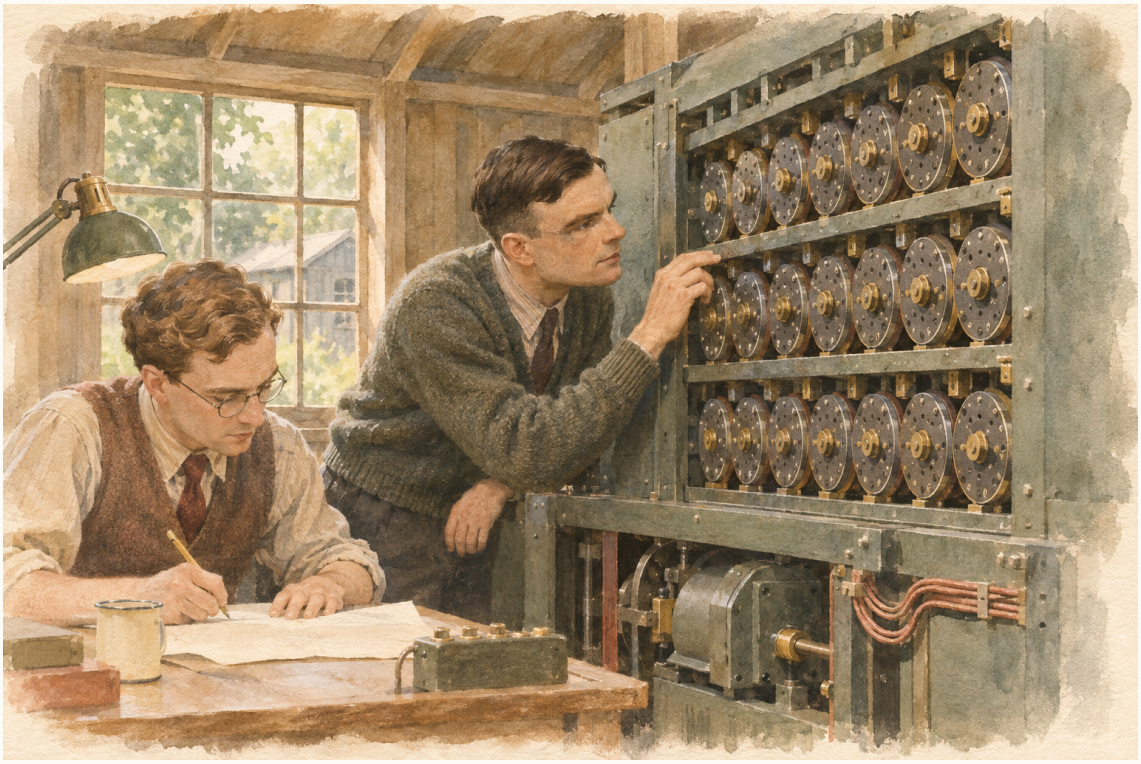
reshaped v.
重塑了

文章简述

- 常态：同一群体的雄性座头鲸会唱同一个版本的歌，通常每年只作细微改变，歌唱可能用于求偶。
- 换歌：研究者在澳洲东部录到的82头鲸中，最初仅两头唱全新歌；约两年后，几乎整个鲸群都改唱新歌。
- 来源：新歌与澳洲西部鲸群的歌相似，研究推测是少量来访鲸把它带到了东部。
- 传播：六个南太平洋鲸群的十余年录音显示，新歌多从澳洲东部出现，再逐群向东传播，通常约两年传到法属波利尼西亚。
- 解释：变化快得来不及靠基因一代代改变，支持鲸会相互模仿、学习的解释；为何换歌、是否等同人类文化，仍有争议。

为机器点燃思想的人

The Man Who Gave Machines a Mind



图灵的贡献，为何没能保护他？

英文原文

Born in June 1912, Alan Turing had a **formidable intellect** by his early twenties. In a paper read in November 1936, he **outlined** a "universal machine": a **device** able to **execute** any **computation**, given the right instructions. That **abstract construct**, **rooted in mathematical logic**, became the **theoretical blueprint** of the modern computer.

On 4 September 1939 he joined the **classified codebreaking effort** at Bletchley Park, leading work on **naval** Enigma, Germany's **encryption** machine. Polish mathematicians had **deciphered** Enigma by late 1932 and shared their **insights** with Britain and France in July 1939. His team's machines **exploited** them and, **operational** from March 1940, **automated** the **laborious** search for each day's **rotor** settings.

In October 1950, writing in the **philosophy** journal Mind, he **posed** a **provocative** question: "Can machines think?" He **proposed** a **practical substitute**, the imitation game, now **widely** called the Turing test. An **interrogator** exchanges typed messages with two **concealed** players, one human and one machine. If the interrogator cannot **consistently distinguish** them, the machine passes.

His **exceptional wartime contribution** did not protect him. On 31 March 1952 he was **convicted of** "gross indecency", the charge then used against gay men. He **forfeited** his **security clearance** and, to **avert imprisonment**, **submitted to hormone injections**. He died of **cyanide poisoning** on 7 June 1954. A **coroner**'s inquest ruled it **suicide**, though some doubt it.

Recognition lagged. On 10 September 2009 Britain's prime minister **publicly apologised**, and on 24 December 2013 Turing **was posthumously granted** a royal **pardon**. On 23 June 2021, his birthday, a fifty-pound note bearing his **portrait** entered **circulation**. The country that **prosecuted** him put his face on its money.

中文译文

艾伦·图灵生于1912年6月，他已拥有强大的才智，二十出头时。在1936年11月宣读的一篇论文中，他勾勒了一种“通用机器”：一种能执行任何计算的装置，只要给出正确的指令。这一抽象的构想根源于数理逻辑，后来成为了现代计算机的理论蓝图。

1939年9月4日，他加入了布莱切利园的保密的破译工作，领导关于海军恩尼格玛密码机的工作，即德国的加密机器。波兰数学家早在1932年底就已破译了恩尼格玛，并将他们的深刻见解（破解思路）于1939年7月告知了英国和法国。他团队（研制的）的机器利用了这些见解和思路，并且从1940年3月起投入运行，自动化了每天搜寻转子设置这项耗时费力的工作。

1950年10月，他在哲学期刊《心灵》（Mind）上撰文，提出了一个发人深省的问题——“机器能思考吗？”他建议用一种切实可行的替代办法取而代之，即模仿游戏，如今普遍称为图灵测试。一名审问者通过打字，与两个隐藏起来的参与者交流，其中一个是人，一个是机器。如果审问者无法始终如一地区分出二者，机器就算通过测试。

他杰出的战时贡献，并没有保护到他。1952年3月31日，他被判定犯有“严重猥亵”罪，这是当时用来惩治男同性恋者的罪名。他被剥夺/丧失了他的安全许可，并且为了避免入狱，被迫接受了激素注射。他死于氰化物中毒，1954年6月7日。验尸官的死因调查裁定为自杀，不过有人对此存疑。

（对于图灵的）认可滞后了。2009年9月10日，英国首相公开道歉了；2013年12月24日，图灵在死后被授予了皇室赦免。2021年6月23日，也就是他的生日，印有他肖像的五十英镑钞票开始流通。起诉他的国家，把他的头像印上了自己的钞票。

词汇与表达

formidable adj.
强大的

intellect n.
才智
a formidable intellect

outlined v.
勾勒了

device n.
装置

execute v.
执行
able to execute any computation

computation n.
计算
able to execute any computation

abstract adj.
抽象的
That abstract construct

construct n.
构想
That abstract construct

rooted in v.
根源于

mathematical adj.
数理

logic n.
逻辑

theoretical adj.
理论

blueprint n.
蓝图
the theoretical blueprint

classified adj.
保密的
the classified codebreaking effort

codebreaking effort n.
破译工作

naval adj.
海军

encryption n.
加密
Germany's encryption machine

deciphered v.
破译了
had deciphered Enigma

词汇与表达（续）

insights n.
深刻见解
shared their insights

exploited v.
利用了

operational adj.
投入运行
operational from March 1940

automated v.
自动化了
automated the laborious search

laborious adj.
耗时费力的
automated the laborious search

rotor n.
转子

philosophy n.
哲学

posed v.
提出了

provocative adj.
发人深省的
a provocative question

proposed v.
建议

practical adj.
切实可行的

substitute n.
替代办法
a practical substitute

widely adv.
普遍

interrogator n.
审问者

concealed adj.
隐藏起来的

consistently adv.
始终如一地
cannot consistently distinguish them

distinguish v.
区分出
cannot consistently distinguish them

exceptional adj.
杰出的

wartime n.
战时

contribution n.
贡献

词汇与表达（续）

convicted of v.
被判定犯有……罪
he was convicted of

forfeited v.
被剥夺/丧失了
He forfeited his security clearance

security clearance n.
安全许可

avert v.
避免
to avert imprisonment

imprisonment n.
入狱

submitted to v.
被迫接受了

hormone n.
激素

injections n.
注射

cyanide poisoning n.
氰化物中毒

coroner n.
验尸官

suicide n.
自杀

Recognition n.
认可

lagged v.
滞后了

publicly adv.
公开

apologised v.
道歉了

was granted v.
被授予了

posthumously adv.
死后
posthumously granted

pardon n.
赦免
a royal pardon

portrait n.
肖像

circulation n.
流通

prosecuted v.
起诉

文章简述

- 理论：1936年，图灵提出按指令执行各种计算的通用机器模型，为现代计算机奠定理论基础。
- 破译：图灵团队借鉴波兰的破译成果，1940年用机器自动搜索德国海军密码机每天的设置，加速破译。
- 测试：1950年，他提出模仿游戏：通过打字交谈，负责问话的人若不能稳定分清人和机器，机器就通过测试。
- 迫害：1952年图灵因同性恋行为获罪，被取消涉密资格，为免入狱接受激素注射；1954年死于氰化物中毒。
- 追认：其死被判为自杀，但有人质疑。英国2009年道歉、2013年给予身后赦免；2021年肖像登上50镑钞票。

没有到来的零日

The Day Zero That Never Came

开普敦怎样避开了水龙头大范围关闭的零日？

英文原文

In January 2018, Cape Town **issued** a **stark alert**: Day Zero could arrive in April. The term **denoted** a **grim threshold**, reached when water in the city's **reservoirs**, as a **percentage** of **capacity**, fell to 13.5 per cent. Most taps would then be shut, and **residents** would queue at about 200 **designated** collection points.

The **prospect** was **genuine**. Three **consecutive** years of **meagre** rainfall, from 2015 to 2017, had **diminished** the reservoirs serving a **metropolitan** area of about four million people. By early 2018 the **dams** were **roughly** a fifth full. The **countdown** turned a **chronic shortage** into an **imminent** deadline.

The city paired **mandatory restrictions** with **unprecedented transparency**. From 1 February 2018, each person was **allotted** 50 **litres** a day. **Officials disclosed** dam levels and **household consumption**; the published **tallies** made **thrift visible** and let neighbours **benchmark** their use. Daily use was more than **halved**, from about 1.2 billion litres in 2015 to about 500 million by early 2018.

Other forces **intervened**. In February 2018, **upstream** farmers released water from their **irrigation** dams; with farms drawing less, Day Zero was **postponed** to July. In March 2018, officials judged Day Zero **unlikely** that year. Winter rains **subsequently replenished** the reservoirs, which were about half full by July. The 50-litre limit was **eased** in October 2018.

The **burden**, critics argued, fell **unevenly**. **Townships**, poor areas **segregated** by race under **apartheid**, had long made do with **scant** water, yet **steeper tariffs squeezed** them hardest. Others blamed **alarmist** warnings for tourist **cancellations**. Households, farmers, and rain kept the collection points **idle**, but the **sacrifice** weighed heaviest on those with the least to give.

中文译文

2018年1月，开普敦发出了一则严峻的警报，称零日可能在4月到来。这个词指的是一道严酷的门槛：即当该市水库的存水按占水库容量的百分比降至13.5%时达到的门槛。大部分水龙头届时将被关闭，居民要到约200个指定的取水点排队取水。

这一前景是真实存在的。连续三年贫乏的降雨，从2015到2017年，减少了水库的蓄水，这些水库给一个大都市区供水，大约400万人口。到2018年初，水库只剩约五分之一的蓄水。倒计时把一场长期的短缺变成了一个迫在眉睫的最后期限。

该市把强制性的限制措施与前所未有的透明度结合起来。自2018年2月1日起，每人被分配每天50升水。官员们公布水库水位和家庭消耗量；公开的统计数字让节俭成效一目了然，也让邻里之间得以对标他们的用水量。日常的用水量减少了一半以上，从2015年的约12亿升降至2018年初的约5亿升。

其他力量也介入了。2018年2月，上游的农民从他们的灌溉水坝中放出了水；加上农场用水减少，零日被推迟到7月。2018年3月，官员们判断零日不太可能在当年发生。冬雨随后重新装满了各水库，到7月时水库已约半满。50升的限额于2018年10月放宽。

批评者认为，负担分配得并不均衡。城镇区，在种族隔离制度下按种族被隔离出来的贫困地区，长期以来就靠很少的水勉强对付，然而更高的水价最大程度地压榨了他们。另一些人则责备危言耸听的警告，因为这些警告造成了游客取消行程。家庭、农民和雨水让取水点一直闲置，但这份牺牲最沉重地落在了那些最没有能力付出的人身上。

词汇与表达

issued v.
发出了
Cape Town issued a stark alert

stark adj.
严峻的
a stark alert

alert n.
警报

denoted v.
指的是
The term denoted a grim threshold

grim adj.
严酷的

threshold n.
门槛
a grim threshold

reservoirs n.
水库
the city's reservoirs

percentage n.
百分比

capacity n.
容量
as a percentage of capacity

residents n.
居民

designated adj.
指定的
200 designated collection points

prospect n.
前景

genuine adj.
真实存在的

consecutive adj.
连续
Three consecutive years

meagre adj.
贫乏的
meagre rainfall

diminished v.
减少了

metropolitan adj.
大都市

dams n.
水库

roughly adv.
约

词汇与表达（续）

countdown n.
倒计时

chronic adj.
长期的
a chronic shortage

shortage n.
短缺

imminent adj.
迫在眉睫的
an imminent deadline

mandatory adj.
强制性的
mandatory restrictions

restrictions n.
限制措施

unprecedented adj.
前所未有的
unprecedented transparency

transparency n.
透明度
unprecedented transparency

allotted v.
分配
each person was allotted 50 litres

litres n.
升

Officials n.
官员们

disclosed v.
公布

household adj.
家庭

consumption n.
消耗量
household consumption

tallies n.
统计数字

thrift n.
节俭

visible adj.
一目了然

benchmark v.
对标
benchmark their use

halved v.
减少了一半

词汇与表达（续）

intervened v.
介入了

upstream adj.
上游的

irrigation n.
灌溉

postponed v.
推迟
Day Zero was postponed to July

unlikely adj.
不太可能

subsequently adv.
随后

replenished v.
重新装满了
replenished the reservoirs

eased v.
放宽

burden n.
负担
The burden, critics argued

unevenly adv.
并不均衡
fell unevenly

Townships n.
城镇区

segregated v.
隔离

apartheid n.
种族隔离制度

scant adj.
很少的

steeper adj.
更高的

tariffs n.
水价
steeper tariffs squeezed them hardest

squeezed v.
压榨了

alarmist adj.
危言耸听的

cancellations n.
取消行程

idle adj.
闲置

sacrifice v.
牺牲

文章简述

- 危机：连续三年少雨使南非开普敦水库告急。2018年初，当局警告，四月可能迎来大范围停水的“零日”。
- 门槛：零日指蓄水降至总容量的13.5%，届时大部分水龙头关闭，居民必须到指定地点领水。
- 节水：全市日用水量由2015年约12亿升降至2018年初约5亿升；2月起每人每日限用50升，并公开水位及用量。
- 转机：居民节水、上游农民放水，以及农场减少取水，使零日一再推迟；随后冬雨补充水库，避免了当年大范围停水。
- 负担：贫困社区原本用水就少，却最难承受上涨水价。批评者指出，危机虽被避开，节水代价并没有被公平分担。

改变人类命运的霉菌

The Mould That Changed the World

从一团霉菌到救命药，中间还差什么？

英文原文

In September 1928 Alexander Fleming, a bacteriologist at St Mary's Hospital, London, returned from holiday to find that a **stray mould** had **contaminated** a dish of staphylococcus **bacteria**. Nothing grew around it. **Popular mythology** blames an open window; the **spores** probably **originated in** a laboratory downstairs, where a colleague **cultivated** moulds.

Fleming called the mould's juice **penicillin** and published in June 1929. Yet the **crude** juice was **unstable** and **stubbornly** hard to **purify**, and he and the chemists he enlisted lacked the **expertise** to **refine** it. He used it **chiefly** to **isolate** other **microbes**, and penicillin **lay dormant**, a **curiosity** rather than a **cure**.

A decade later, in Oxford, Howard Florey, an Australian pathologist, **revived** the work with Ernst Chain, a German-born **chemist** and Jewish **refugee**. Their colleague Norman Heatley **improvised** an **ingenious apparatus** to **extract** the **compound**. On 25 May 1940 eight mice received a **lethal dose** of bacteria; only the four also **injected with** penicillin lived. The Lancet, a **prestigious** journal, printed the **outcome** that August.

In February 1941 Albert Alexander, an Oxford **constable gravely ill** with **infection**, **rallied** on penicillin. When the **scarce** supply **ran out**, he died in March: the **obstacle** was **quantity**. From July 1941 a government laboratory in Peoria, Illinois, **boosted** yields about **tenfold** with **corn-steep liquor**, a **starch** by-product; later a **mouldy** market **melon furnished** the laboratory **with** a more **prolific** strain. By D-Day in June 1944 **industrial fermentation** met **Allied** needs.

In October 1945 Fleming, Florey, and Chain shared the Nobel prize; Heatley was **overlooked**. **Fortune** put the mould in Fleming's dish; **tenacious** teams and years of **toil transformed** it **into** a **remedy**.

中文译文

1928年9月，亚历山大·弗莱明，伦敦圣玛丽医院的细菌学家，度假归来，发现一团零星的/散落的霉菌污染了一只培养葡萄球菌这种细菌的培养皿。它周围什么也没长。坊间传说将其归咎于一扇敞开的窗户；孢子很可能起源于楼下的一间实验室，在那里有一位同事培育了霉菌。

弗莱明将这种霉菌（分泌的）汁液称为青霉素，并于1929年6月发表。然而，这种粗糙的/未加工的液体很不稳定，并且极难提纯；他和招募来的化学家都缺乏精炼它的专业知识。他主要用它来分离其他微生物，青霉素就此休眠，成了一种稀罕物/奇物，而非疗法。

约十年后，在牛津，霍华德·弗洛里，一位澳大利亚病理学家，与恩斯特·钱恩一起重启了这项研究，钱恩是德国出生的化学家、犹太难民。他们的同事诺曼·希特利用手边的材料临时拼凑了一套巧妙的装置，用来提取这种化合物。1940年5月25日，八只小鼠接种了致死剂量的细菌；只有另外注射了青霉素的那四只活了下来。享有盛誉的期刊《柳叶刀》在同年8月刊登了这一结果。

1941年2月，艾伯特·亚历山大，一名牛津警员，因感染而病情危重，用上青霉素后有所好转。当稀缺的（青霉素）供应耗尽后，他于3月去世：障碍在于数量。从1941年7月起，伊利诺伊州皮奥里亚的一家政府实验室把产量提高了约十倍，靠的是玉米浆，一种淀粉生产的副产品；后来，市场上一只发霉的甜瓜又为这家实验室提供了一株更高产的菌株。到1944年6月诺曼底登陆时，工业发酵已能满足盟军的需求。

1945年10月，弗莱明、弗洛里和钱恩共同获得了诺贝尔奖；希特利却遭到忽视。机缘把霉菌送进了弗莱明的培养皿；几支锲而不舍的/坚持不懈的团队加上多年的辛劳/辛苦工作，把它转化成了良药。

词汇与表达

stray adj.
零星的/散落的

mould n.
霉菌

contaminated v.
污染了
a stray mould had contaminated

bacteria n.
细菌

Popular mythology n.
坊间传说

spores n.
孢子
the spores probably originated

originated in v.
起源于

cultivated v.
培育了

penicillin n.
青霉素
the mould's juice penicillin

crude adj.
粗糙的/未加工的
the crude juice

unstable adj.
不稳定
the crude juice was unstable

stubbornly adv.
极

purify v.
提纯
stubbornly hard to purify

expertise n.
专业知识
lacked the expertise to refine it

refine v.
精炼

chiefly adv.
主要

isolate v.
分离
to isolate other microbes

microbes n.
微生物

lay dormant adj.
休眠

curiosity n.
稀罕物/奇物

词汇与表达（续）

cure n.
疗法

revived v.
重启了
revived the work

chemist n.
化学家

refugee n.
难民

improvised v.
临时拼凑了
improvised an ingenious apparatus

ingenious adj.
巧妙的
an ingenious apparatus

apparatus n.
装置
an ingenious apparatus

extract v.
提取
to extract the compound

compound n.
化合物

lethal adj.
致死
a lethal dose of bacteria

dose n.
剂量

injected with v.
注射了

prestigious adj.
享有盛誉的

outcome n.
结果

constable n.
警员

gravely ill adv.
病情危重

infection n.
感染

rallied v.
有所好转

scarce adj.
稀缺的
the scarce supply

词汇与表达（续）

ran out 动词短语
耗尽

obstacle n.
障碍
the obstacle was quantity

quantity n.
数量
the obstacle was quantity

boosted v.
提高了

tenfold adv.
十倍
boosted yields about tenfold

corn-steep liquor 名词短语
玉米浆

starch n.
淀粉

mouldy n.
发霉的

melon n.
甜瓜

furnished with v.
为……提供了

prolific adj.
高产的
a more prolific strain

industrial adj.
工业

fermentation n.
发酵
industrial fermentation

Allied adj.
盟军的

overlooked v.
遭到忽视

Fortune n.
机缘

tenacious adj.
锲而不舍的/坚持不懈的
tenacious teams

toil n.
辛劳/辛苦工作

transformed into v.
转化成了

remedy n.
良药
transformed it into a remedy

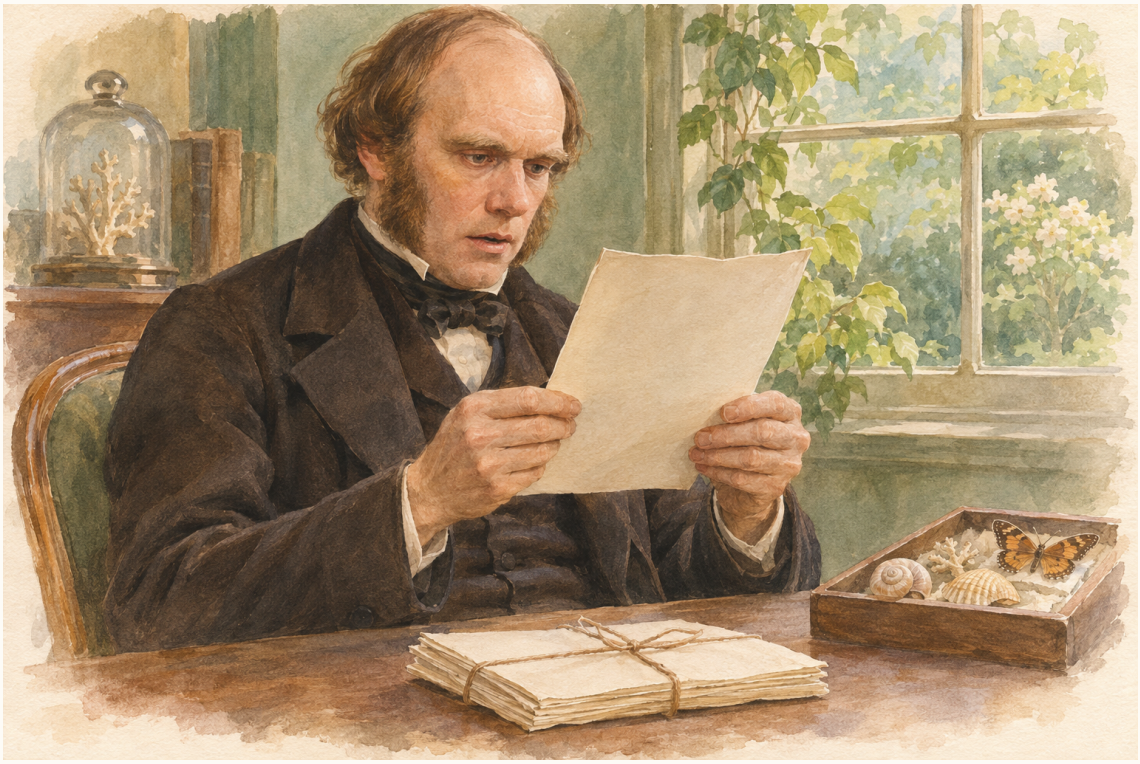
文章简述

- 偶然：1928年，弗莱明发现培养皿里的霉菌能抑制细菌，将霉菌汁称作青霉素；霉菌孢子可能来自楼下实验室，从窗外飘来的说法只是传闻。
- 难关：粗制青霉素不稳定，又难从混合液中分离出有效成分，长期未能成为药物；约十年后，弗洛里、钱恩和希特利等人接力，改进提取方法。
- 验证：8只感染小鼠中，只有同时用了青霉素的4只存活；1941年一名重病警员用药后一度好转，却因药物用尽而死亡，说明产量不足仍是难关。
- 量产：美国研究者用玉米浆把产量提高约十倍，发霉甜瓜又提供了高产菌株；到1944年，工业生产已能满足盟军需要。
- 归功：1945年弗莱明、弗洛里和钱恩获诺奖，希特利未入列；一次偶然发现，经过多支团队多年努力，才变成救人的药。

65

一封信惊醒了达尔文

The Letter That Startled Darwin



华莱士的来信为何让达尔文不安？

英文原文

Charles Darwin **disembarked** from HMS Beagle in October 1836. In September 1838 he began **formulating** the **principle** of natural **selection**: individuals better **adapted to** their **surroundings** survive and **pass on** their **hereditary traits**. By 1844 he had **composed** a 230-page **manuscript**, yet kept it **unpublished**, **cautiously amassing** evidence.

In February 1858, two decades after Darwin's first insight, Alfred Russel Wallace, a **naturalist afflicted by** fever in the Malay Archipelago, **independently conceived** the same idea. His **concise** essay reached Darwin on 18 June 1858. Its argument **paralleled** Darwin's own almost **precisely**; he told Charles Lyell he had never seen a more **remarkable coincidence**. His claim to **priority** seemed **threatened**.

Lyell and the **botanist** Joseph Hooker **arranged for** both papers to be read **simultaneously** at the Linnean Society, a scientific **institution** in London, on 1 July 1858. Neither author attended the **proceedings**. Wallace was abroad, and Darwin was **grieving**: **scarlet** fever had killed his **infant** son on 28 June. The **compromise**, struck without Wallace, may have kept **rivalry** over **precedence** at bay.

The papers' **publication** on 20 August 1858 met a **muted reception**. In May 1859 the Society's president **remarked** in his **yearly** address that the year had been **barren of revolutionary** discoveries. The **sensation** came on 24 November 1859 with Darwin's **treatise**, **On the Origin of Species**: booksellers had **subscribed for** every copy of its first **impression**, 1,250 in all.

Wallace had **readily signified** his **acceptance** in October 1858, **assuring** Hooker in a **cordial** letter of his **satisfaction** with the **arrangement**. In 1889 he published a **lengthy exposition** of the idea and **entitled** it Darwinism. Wallace never **named** the theory **after** himself.

中文译文

1836年10月，查尔斯·达尔文从小猎犬号上登岸。1838年9月，他开始构想并明确表述自然选择的原理：对周围环境适应得更好的个体存活下来，并将传递他们自身遗传的性状（给后代）。到1844年，他已撰写了一份230页的手稿，却一直未予发表，谨慎地积累着证据。

1858年2月，达尔文最初的洞察出现后的二十年，博物学家阿尔弗雷德·拉塞尔·华莱士在马来群岛饱受热病的折磨，独自一人构想出了同一个想法。华莱士这篇简明的文章于1858年6月18日到达了达尔文的手里。文中的论证与达尔文自己的几乎完全一样；达尔文告诉查尔斯·莱尔，自己从未见过比这更惊人的巧合。他对优先权的主张似乎受到了威胁。

莱尔和植物学家约瑟夫·胡克安排这两篇论文，于1858年7月1日在伦敦的科学机构林奈学会上被同时宣读。两位作者都没有出席这场会议。华莱士身在海外，达尔文则处于哀痛之中：猩红热在6月28日夺去了他幼子的生命。这一折中办法在华莱士未参与的情况下达成，或许避免了围绕先后次序的竞争。

这些论文于1858年8月20日的发表，遭遇了一个很不热烈的反响。1859年5月，学会会长在年度致辞中评论说，这一年一直毫无革命性的发现。这场轰动于1859年11月24日随着达尔文的专著《物种起源》到来：书商们已订购了该书首次印刷的每一册，总计1,250册。

华莱士早已欣然表明了接受的态度，那是在1858年10月；他在一封亲切的信中向胡克保证，自己对这一安排感到满意。1889年，他出版了一部长篇的阐述解释这一思想，并将其命名为《达尔文主义》。华莱士从未用自己的名字为这一理论命名。

词汇与表达

disembarked v.
登岸

formulating v.
构想并明确表述
he began formulating the principle of

principle n.
原理
he began formulating the principle of

selection n.
选择

adapted to v.
对……适应
individuals better adapted to their

surroundings n.
周围环境

pass on v.
传递

hereditary adj.
遗传的
and pass on their hereditary traits.

traits n.
性状
and pass on their hereditary traits.

composed v.
撰写了

manuscript n.
手稿
he had composed a 230-page manuscript,

unpublished adj.
未予发表

cautiously adv.
谨慎地

amassing v.
积累着
it unpublished, cautiously amassing

naturalist n.
博物学家

afflicted by v.
饱受……的折磨
a naturalist afflicted by fever in the

independently adv.
独自一人
independently conceived the same idea.

conceived v.
构想出
independently conceived the same idea.

concise adj.
简明的
His concise essay reached Darwin on 18

paralleled v.
与……一样

precisely adv.
完全

remarkable adj.
惊人的

coincidence n.
巧合
seen a more remarkable coincidence.

priority n.
优先权
claim to priority seemed threatened.

threatened v.
受到了威胁

botanist n.
植物学家

arranged for v.
安排

simultaneously adv.
同时
to be read simultaneously at the

词汇与表达（续）

institution n.
机构

proceedings n.
会议

grieving v.
哀痛

scarlet adj.
猩红

infant adj.
幼子

compromise n.
折中办法
compromise, struck without Wallace,

rivalry n.
竞争
may have kept rivalry over precedence

precedence n.
先后次序
may have kept rivalry over precedence

publication n.
发表

muted adj.
很不热烈的
20 August 1858 met a muted reception.

reception n.
反响
20 August 1858 met a muted reception.

remarked v.
评论

yearly adj.
年度

barren of adj.
毫无

revolutionary adj.
革命性的

sensation n.
轰动

On the Origin of Species n.
《物种起源》

treatise n.
专著

subscribed for v.
订购了

impression n.
次印刷

readily adv.
欣然

signified v.
表明了

acceptance n.
接受

assuring v.
向……保证

cordial adj.
亲切的
a cordial letter of his satisfaction

satisfaction n.
满意

arrangement n.
安排

lengthy adj.
长篇的

exposition n.
阐述
a lengthy exposition of the idea and

entitled v.
将……命名为
exposition of the idea and entitled it

named after v.
用……为……命名

文章简述

- 酝酿：达尔文1838年构想自然选择：更适应环境的生物较容易存活，并把自身特征传给后代。1844年他已写成230页手稿，仍继续搜集证据，没有发表。
- 来信：华莱士独立想出同一原理，1858年寄来的文章几乎与达尔文一致，令达尔文担心自己的首创地位。
- 折中：莱尔、胡克安排两人成果于1858年7月同场宣读；两位作者都未出席，华莱士也未参与商定办法。
- 反响：论文发表初期反响冷淡；1859年《物种起源》引起轰动，首印1250册已被书商全部订购。
- 认可：华莱士1858年10月事后致信认可安排；1889年又以《达尔文主义》阐述自然选择，未用自己名字命名理论。

入侵澳洲的毒蟾

The Toad That Invaded Australia

引进的灭虫帮手，为什么成了生态难题？

英文原文

Greyback cane **beetles**, whose **larvae devour** roots, were **plaguing lucrative** sugar **plantations** in Queensland, north-eastern Australia. In June 1935 Queensland **imported** 102 cane toads, **bulky amphibians native to** the Americas, from Hawaii. That August about 2,400 of their young were **released**. **Biological** control, setting one **species** against a **pest**, had an **ambitious** new **recruit**.

The **hasty** plan met **opposition**. In November 1935 an insect expert, Walter Froggatt, **persuaded** officials to **halt** releases; after Queensland's **lobbying**, they **resumed** in September 1936. The **scheme flopped**: the toads barely **dented** beetle numbers, perhaps because the insects stayed high up on cane **stalks**.

The toads **thrived** instead. With few **foes** and **fertile** females **spawning** thousands of eggs, they **multiplied unchecked**. Their **invasion frontier**, creeping about ten kilometres a year at first, **accelerated** to about 50 a year in the **tropical** north. In February 2009 the first known toad **breached** Western Australia's border.

Toxin glands in their skin make them a deadly meal. Numbers of goannas (large **lizards**) and northern quolls (small **marsupial** hunters) have **slumped** in places; some local quoll populations were **wiped out**. The toads have **wreaked havoc**, but the damage is **uneven**; some species have **prospered**.

Biological control can **triumph**. From 1926 the cactus **moth**, Cactoblastis, cleared most **prickly** pear, an **invasive cactus**, from Queensland and New South Wales within about a decade. Tested first, its **larvae starved** rather than eat fifteen other plants; the toads faced no such test. Under an October 1984 law, recruits now face **stringent scrutiny**. The **blunder** was born of **haste** and sealed by **lobbying** that **overrode** a warning: an **untested ally**, once **loose**, may never be **recalled**.

中文译文

灰背蔗甲虫它的幼虫啃食根系，正困扰着利润丰厚的甘蔗种植园，此种植园位于澳大利亚东北部的昆士兰州。1935年6月，昆士兰州从夏威夷引进了102只甘蔗蟾蜍，这是一种原产于美洲、庞大笨重的两栖动物。同年8月，它们的约2,400只后代被释放到野外。生物的防治，即用一个物种去对付一种害虫，就此迎来了一名雄心勃勃的新兵。

这个草率的计划遭到了反对。1935年11月，昆虫专家沃尔特·弗罗格特说服官员停止释放；经昆士兰州游说，释放于1936年9月恢复。这套计划/方案彻底失败：蟾蜍几乎没能削减甲虫的数量，或许是因为这些昆虫一直待在甘蔗茎秆的高处。

蟾蜍自己反倒兴旺起来。天敌寥寥，能生育的雌蟾产下数千枚卵，它们不受遏制地大量繁殖。它们的入侵前沿起初每年缓慢行进约十公里，到了热带北部，加快到每年约50公里。2009年2月，已知的第一只蟾蜍突破了西澳大利亚州的边界。

它们皮肤里的毒素腺体，使它们成为了一顿致命的餐食。巨蜥（大型蜥蜴）和北袋鼬（一种小型有袋类猎手）的数量在一些地方锐减；部分地方的袋鼬种群被彻底消灭。蟾蜍造成了严重破坏，但损害并不均衡；有些物种反而繁荣了起来。

生物防治也能大获成功。从1926年起，仙人掌螟这种蛾子在约十年内清除了大部分“带刺的梨”，这是一种外来入侵的仙人掌，清除范围遍及昆士兰州和新南威尔士州。这种蛾子放出前经过检验，其幼虫宁可饿死也不吃另外十五种植物；蟾蜍却从未受过这样的检验。根据1984年10月通过的一部法律，如今新兵都要经过严格的认真彻底的审查。这个大错起于仓促，又因游说压过了警告而成定局：未经检验的盟友一旦放出，就可能再也召不回来。

词汇与表达

beetles n.
甲虫

larvae n.
幼虫
whose larvae devour roots

devour v.
啃食
larvae devour roots

plaguing v.
困扰

lucrative adj.
利润丰厚的
lucrative sugar plantations

plantations n.
种植园

imported v.
引进了

bulky adj.
庞大笨重的

amphibians n.
两栖动物
bulky amphibians

native to phr.
原产于
native to the Americas

released v.
释放

Biological adj.
生物的

species n.
物种
setting one species against a pest

pest n.
害虫
one species against a pest

ambitious adj.
雄心勃勃的

recruit n.
新兵

hasty adj.
草率的
The hasty plan met opposition

opposition n.
反对
The hasty plan met opposition

persuaded v.
说服
persuaded officials to halt releases

halt v.
停止

词汇与表达（续）

lobbying n.
游说
after Queensland's lobbying

resumed v.
恢复
they resumed in September 1936

scheme n.
计划/方案

flopped v.
彻底失败

dented v.
削减

stalks n.
茎秆

thrived v.
兴旺
The toads thrived instead

foes n.
天敌

fertile adj.
能生育的

spawning v.
产下

multiplied v.
大量繁殖

unchecked adj.
不受遏制地
they multiplied unchecked

invasion n.
入侵
Their invasion frontier

frontier n.
前沿

accelerated v.
加快
accelerated to about 50 a year

tropical adj.
热带

breached v.
突破了

Toxin n.
毒素
Toxin glands in their skin

glands n.
腺体

lizards n.
蜥蜴

marsupial adj.
有袋类

词汇与表达（续）

slumped v.
锐减

wiped out phr.
彻底消灭
quoll populations were wiped out

wreaked havoc phr.
造成了……严重破坏
The toads have wreaked havoc, but the damage is uneven;

uneven adj.
不均衡

prospered v.
繁荣了

triumph v.
大获成功

cactus n.
仙人掌

moth n.
蛾子

prickly adj.
带刺的

invasive adj.
外来入侵的
an invasive cactus

starved v.
饿死

stringent adj.
严格的
stringent scrutiny

scrutiny n.
认真彻底的审查
stringent scrutiny

blunder n.
大错

haste n.
仓促

overrode v.
压过了
lobbying that overrode a warning

untested adj.
未经检验的

ally n.
盟友

loose adj.
放出

recalled v.
召不回来

文章简述

- 引进：1935年，澳大利亚为对付危害甘蔗的甲虫，引进甘蔗蟾蜍，再把约2400只后代放进野外，希望它们吃掉害虫。
- 冒进：专家曾促成暂停释放，但地方游说让计划恢复；放出前却缺少检查蟾蜍究竟吃什么的试验。
- 失控：甲虫没被明显消灭，蟾蜍却因天敌少、繁殖快而不断扩散，在热带北部每年推进约50公里。
- 伤害：蟾蜍皮肤有毒，捕食它们的巨蜥、北袋鼬在一些地方大幅减少，部分袋鼬种群消失；对不同物种的影响并不一致。
- 教训：此前引进仙人掌螟时，试验发现它的幼虫宁可饿死也不吃另外15种植物，后来才用于控制仙人掌；蟾蜍没有经过同类食性检验。未经检验的“帮手”一旦失控，可能再也收不回来。

67

从一道划痕到天花绝迹

From a Scratch to a World Without Smallpox

詹纳不是第一人，贡献究竟在哪里?

英文原文

Smallpox, a **contagious affliction**, killed about three in ten of its **sufferers** and an estimated 300 million in the 20th century. An early **safeguard**, **deliberately implanting matter** from its **sores** in the healthy, was **practised** in China by the 16th century and reached England in 1721. The **procedure** itself could cause **grave sickness** or death, if less **frequently** than natural smallpox.

Rustic lore held that milkmaids with cowpox, a **mild ailment** of **livestock**, were **immune to** smallpox. In 1774 Benjamin Jesty, a Dorset farmer, **employed** cowpox to protect his wife and two sons. On 14 May 1796 Edward Jenner, a **provincial practitioner**, **inserted fluid** from a milkmaid's cowpox sore into a **scratch** on eight-year-old James Phipps.

On 1 July 1796 Jenner **exposed** the boy **to** smallpox; no **symptoms ensued**. In 1797 the Royal Society declined to **publish** his paper, judging his evidence **unconvincing**. In June 1798 he **privately funded** his book, the Inquiry, with **supplementary** cases. The word **vaccine derives from** the **Latin** vacca, cow.

Jenner was not the idea's **originator**. His contribution was a test, with results **accessible** in print, so other doctors could **replicate** and **disseminate** the **technique**. By today's standards, **experimentation** on a child without **informed consent** would be **unethical**. Jenner died on 26 January 1823, with smallpox far from **eliminated**.

In 1967 the WHO launched an **intensified**, **concerted** campaign to **eradicate** smallpox. The last **naturally occurring** case was **diagnosed** in Somalia on 26 October 1977. On 8 May 1980 the World Health Assembly **formally** declared the world free of smallpox, the only human disease ever **wiped out**. That **pronouncement** came almost 184 years after Phipps was **vaccinated**.

中文译文

天花，一种传染性的疾病，夺去了约三成患者的生命，并且估计有3亿人在20世纪丧命。一种早期的防护手段，是有意把天花脓疮中的物质植入健康人体内，到16世纪，中国已在施行这种方法，1721年又传入英国。这种做法本身可能导致严重的疾病或死亡，尽管不如自然天花那样频繁。

乡间传说认为，挤奶女工若得过牛痘这种牲畜的轻微病症，就对天花免疫。1774年，本杰明·杰斯蒂，一位多塞特郡农夫，使用牛痘来保护他的妻子和两个儿子。1796年5月14日，爱德华·詹纳，一位地方执业医师，将取自一名挤奶女工牛痘疮的液体，注入8岁的詹姆斯·菲普斯的一道划痕中。

1796年7月1日，詹纳让这个男孩接触天花；任何症状都没有随之出现。1797年，皇家学会拒绝去发表他的论文，认为他的证据缺乏说服力。1798年6月，他私人出资出版了他的著作《探究》，书中附上了补充病例。表示疫苗的英文词源自拉丁语的vacca，意思是牛。

詹纳并不是这一想法的首创者。他的贡献是一项试验，其结果在已出版的书籍中可以获得，以便其他医生能够复制并传播这项技术。按今天的标准，拿孩子做实验在没有取得知情同意的情况下，属于不道德行为。詹纳于1823年1月26日去世，那时天花还远未被消除。

1967年，世界卫生组织发起了一场强化、共同筹划的运动，要根除天花。最后一例自然发生的病例于1977年10月26日在索马里确诊。1980年5月8日，世界卫生大会正式宣布全世界已摆脱天花，天花也成为迄今唯一被彻底消灭的人类疾病。那一宣告，出现在菲普斯接种疫苗之后的近184年。

词汇与表达

Smallpox n.
天花

contagious adj.
传染性的
Smallpox, a contagious affliction,

affliction n.
疾病
Smallpox, a contagious affliction,

sufferers n.
患者

safeguard n.
防护手段
safeguard, deliberately implanting

deliberately adv.
有意

implanting v.
植入
safeguard, deliberately implanting

matter n.
物质

sores n.
脓疮

practised v.
施行

procedure n.
做法
The procedure itself could cause grave

grave adj.
严重的
itself could cause grave sickness or

sickness n.
疾病

frequently adv.
频繁

Rustic adj.
乡间

lore n.
传说

mild adj.
轻微

ailment n.
病症

livestock n.
牲畜

immune to adj.
对……免疫
of livestock, were immune to smallpox.

employed v.
使用

provincial adj.
地方

practitioner n.
执业医师
Jenner, a provincial practitioner,

inserted v.
注入

fluid n.
液体

scratch n.
划痕

exposed to v.
让……接触

symptoms n.
症状

ensued v.
随之出现
no symptoms ensued.

词汇与表达（续）

publish v.
发表

unconvincing adj.
缺乏说服力

privately adv.
私人

funded v.
出资

supplementary adj.
补充
with supplementary cases.

vaccine n.
疫苗
The word vaccine derives from the

derives from v.
源自

Latin n.
拉丁语

originator n.
首创者
Jenner was not the idea's originator.

accessible adj.
可以获得
was a test, with results accessible in

replicate v.
复制
so other doctors could replicate and

disseminate v.
传播
could replicate and disseminate the

technique n.
技术

experimentation n.
实验

informed adj.
知情
on a child without informed consent

consent n.
同意
on a child without informed consent

unethical adj.
不道德
would be unethical.

eliminated v.
消除

intensified adj.
强化

concerted adj.
共同筹划的
an intensified, concerted campaign to

eradicate v.
根除
campaign to eradicate smallpox.

naturally adv.
自然

occurring v.
发生

diagnosed v.
确诊
was diagnosed in Somalia on 26 October

formally adv.
正式

wiped out phr.
彻底消灭

pronouncement n.
宣告
That pronouncement came almost 184

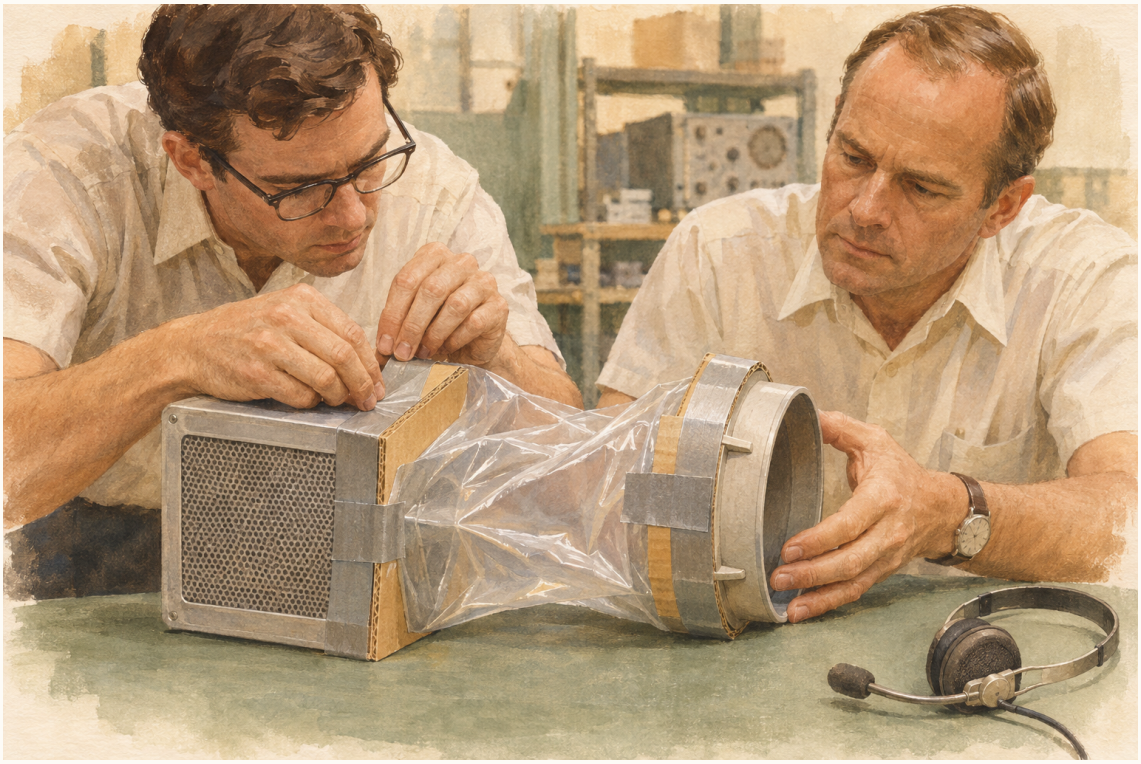
vaccinated v.
接种疫苗

文章简述

- 背景：天花约致三成患者死亡；早期人痘术把天花疮中物质接种给健康人作预防，风险虽低于自然染病，仍可能致重病或死亡。
- 先例：乡间传牛痘可防天花；农夫杰斯蒂1774年已用牛痘保护家人，詹纳并非首创。
- 试验：詹纳1796年给男孩种牛痘，再让他接触天花而未发病；无知情同意的儿童试验按今天标准不合伦理。
- 推广：皇家学会嫌证据不足拒稿后，詹纳自费公布试验和补充病例，使其他医生能够重复验证、传播技术。
- 根除：世卫1967年强化行动，1977年出现最后自然病例；1980年世界卫生大会宣布根除天花，迄今唯一被根除的人类疾病。

阿波罗13号：成功的失败

Apollo 13: The Successful Failure



方形滤芯怎样接进圆形插槽，救回三名宇航员？

英文原文

On April 11th 1970 NASA **launched** Apollo 13, meant for the third **crewed lunar landing**. On April 13th, some 200,000 miles out, an **oxygen tank exploded** in the **service module**, home to most of the ship's oxygen and power. The **blast crippled** the **spacecraft**; the landing was **aborted**.

The **astronauts took refuge in** the **lunar module**, Aquarius, built to support two men for about two days. Using its **engine**, the crew **steered** the **craft onto** a **trajectory** around the Moon, whose gravity **swung** it home. They **rationed** water, **conserved power**, and **shivered** in a **bitterly** cold **cabin**.

A quieter **hazard emerged**: with a third man aboard, **carbon dioxide** was **accumulating**. The command module, the crew's **capsule**, used square **filter cartridges**; Aquarius's **sockets** were round. From plastic bags, **cardboard**, and **tape** on board, ground engineers **devised** an **adapter**, then **talked** the crew **through assembling** it. On April 17th the capsule **splashed down** in the Pacific.

Yet the **fault** had been **built in** on Earth. A NASA **review board traced** the **explosion to** the tank's **thermostat switches**, designed for the ship's 28-**volt** supply. In 1965 the spacecraft's **builder rewrote** the **specification** to let the **heaters run on** 65-volt launch-site power; the switches were never **upgraded**, a **mismatch** neither that **contractor** nor NASA **detected**. Weeks before launch, to **drain** the **stubborn** tank, a **heater** ran about eight hours. The switches probably **malfunctioned**; **unchecked** heat **scorched** the **insulation** on the tank's **wiring**; in flight, that insulation **ignited**.

NASA called Apollo 13 a "successful failure". The **paradox** holds: a flight **undone by** a five-year-old engineering **oversight** was **salvaged** within four days by **ingenuity** on the ground.

中文译文

1970年4月11日，NASA发射了阿波罗13号，原本是要完成第三次载人的登月。4月13日，大约离地球200,000英里远的地方，一个氧气罐爆炸了，发生在服务舱段里面，存储着大部分飞船的氧气和电力。这股爆炸气浪严重毁坏了宇宙飞船；登月被终止了。

宇航员们躲进登月舱水瓶座号避难，它被建造出来原本是为了支持两个人生存大约两天左右。使用它的发动机/引擎，机组操纵航天器驶入一条绕月轨迹，月球的引力把它甩回了地球。他们限定用水量，节约用电，并且不停地打哆嗦，身处极其寒冷的舱室里。

一个悄无声息的隐患出现了：随着第三个人登舱，二氧化碳正在积聚。指令舱，即机组太空舱，用的是方形的过滤器滤芯；然而水瓶座号的插槽却是圆的。用飞船上现有的塑料袋、硬纸板和胶带，地面工程师构思设计出一个转接器/转换插头，然后一步步指导机组把它组装起来。4月17日，太空舱水上降落在太平洋。

然而，这个缺陷早在地球上就被内置了/埋下了。一个NASA的审查委员会查明了这次爆炸源自于（液氧）罐的恒温器开关上，这些开关是按飞船的28伏电源设计的。在1965年，飞船承建商就改写了技术规格，允许加热器使用发射场65伏的电源来运行；开关却始终没有升级，这处不匹配，无论是那家承包商还是NASA都没有发现。发射前几周，为了排空那个难以排空的储罐，一台加热器连续工作了约八个小时。开关很可能失灵了；无人管控的高温烧焦了罐内线路上的绝缘层；在飞行途中，这层绝缘材料燃烧起来了/着火了。

NASA把阿波罗13号称作“成功的失败”。这个悖论站得住脚：一次飞行，毁于五年前的一处工程疏忽，却在四天之内得以挽救，靠的是地面上的巧思/足智多谋。

词汇与表达

launched v.
发射了

crewed adj.
载人的

lunar landing adj.
登月

oxygen tank n.
氧气罐

exploded v.
爆炸了

service module n.
服务舱段

blast n.
爆炸气浪

crippled v.
严重毁坏了
The blast crippled the spacecraft

spacecraft n.
宇宙飞船

aborted v.
终止了
the landing was aborted

astronauts n.
宇航员

took refuge in n.
躲进……避难
The astronauts took refuge in the lunar module, Aquarius,

lunar module adj.
登月舱

engine n.
发动机/引擎

steered onto v.
操纵……驶入

craft n.
航天器

trajectory n.
轨迹
a trajectory around the Moon

swung v.
甩回了

rationed v.
限定……量
They rationed water

conserved power v.
节约用电

shivered v.
打哆嗦

bitterly adv.
极其

cabin n.
舱室

hazard n.
隐患
A quieter hazard emerged

词汇与表达（续）

emerged v.
出现了
A quieter hazard emerged

carbon dioxide n.
二氧化碳

accumulating v.
积聚
carbon dioxide was accumulating

capsule n.
太空舱

filter n.
过滤器

cartridges n.
滤芯

sockets n.
插槽

cardboard n.
硬纸板

tape n.
胶带

devised v.
构思设计出
ground engineers devised an adapter

adapter n.
转接器/转换插头
devised an adapter

talked through v.
一步步指导

assembling v.
组装
through assembling it

splashed down v.
水上降落

fault n.
缺陷
the fault had been built in on Earth

built in v.
内置了/埋下了

review board n.
审查委员会

traced to v.
查明了……源自于

explosion n.
爆炸

thermostat n.
恒温器
the tank's thermostat switches

switches n.
开关

volt n.
伏

词汇与表达（续）

builder n.
承建商

rewrote v.
改写了

specification n.
技术规格
rewrote the specification

heaters n.
加热器

run on v.
使用……的电源来运行

upgraded v.
升级

mismatch n.
不匹配
a mismatch

contractor n.
承包商

detected v.
发现

drain v.
排空
to drain the stubborn tank

stubborn adj.
难以排空的

heater n.
加热器

malfunctioned v.
失灵了
The switches probably malfunctioned

unchecked adj.
无人管控的

scorched v.
烧焦了

insulation n.
绝缘层
the insulation on the tank's wiring

wiring n.
线路

ignited v.
燃烧起来了/着火了

paradox n.
悖论
The paradox holds

undone by v.
毁于

oversight n.
疏忽
a five-year-old engineering oversight

salvaged v.
挽救
was salvaged within four days

ingenuity n.
巧思/足智多谋
by ingenuity on the ground

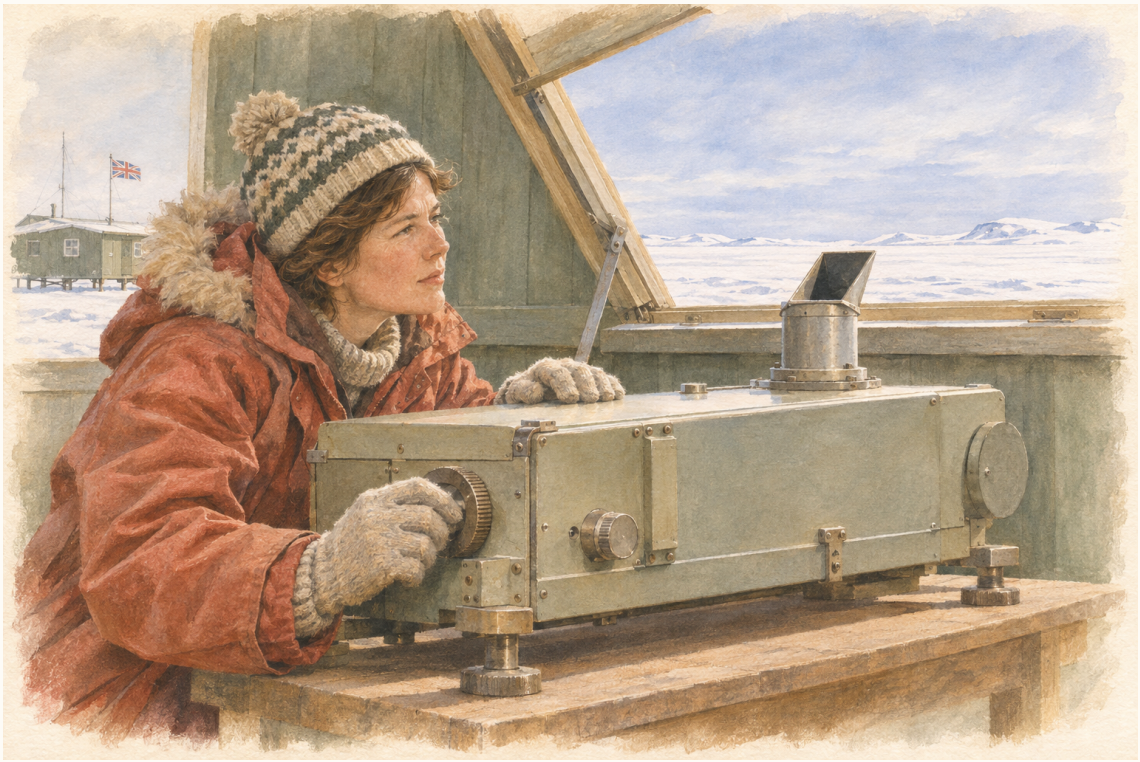
文章简述

- 事故：1970年，阿波罗13号服务舱的氧气罐爆炸，登月取消。三名宇航员躲进原本只供两人使用约两天的登月舱。
- 险情：他们缺水缺电、忍受严寒，舱内二氧化碳还不断积聚；能清除它的方形滤芯，却接不上登月舱的圆形接口。
- 救急：地面工程师用船上已有的塑料袋、纸板和胶带设计转接器，再指导机组装好，解决滤芯不匹配的问题。
- 根源：氧气罐加热器的规格已改为兼容发射场的65伏电源，罐内温控开关却仍按飞船的28伏电源设计。发射前长时间加热时，开关可能失灵，烧伤电线绝缘层，后来在飞行中起火。
- 返航：机组用登月舱发动机调整路线，借月球引力绕回地球，爆炸四天后安全溅落。没登成月，却救回三人，故称“成功的失败”。

69

拯救天空

Saving the Sky



臭氧层的低值，为什么被软件怀疑?

英文原文

A **layer** of **ozone shields** Earth **from** much of the sun's **ultraviolet radiation**. In June 1974 Mario Molina and Sherwood Rowland, chemists in California, warned in Nature of a **menace** in fridges and **spray** cans. Chlorofluorocarbons (CFCs) are so **inert** that they drift **intact** into the stratosphere, high in the **atmosphere**, where they can **liberate chlorine**, a **catalyst** that destroys ozone **molecule** after molecule. The pair foresaw **gradual erosion**.

The **alarm**, when it sounded, was **abrupt**. In May 1985 Joseph Farman and colleagues at the British **Antarctic** Survey reported that **ozone** over Halley Bay, their Antarctic station, had **plummeted** each southern spring since the mid-1970s. Their **humble** Dobson spectrophotometer **gauged** ozone from the ground. The staff were **persistent**, reading it in a **painstaking routine** of **manual observation**.

Yet NASA's Nimbus-7 **satellite**, a **sophisticated** eye **in orbit**, had recorded the same lows. Its software **flagged** them **as dubious**, since ozone that low had never been reliably measured. NASA says it was still **verifying** them when Farman published. The **sensor** had worked; **expectation** had not.

Remarkably, the response was **swift**. In September 1987 **diplomats** signed the Montreal **Protocol**, a **binding accord** to **curb** CFCs. **Amendments tightened** it: in June 1990 **affluent** countries **pledged** to **phase out** CFCs by 2000. In September 2009 the Protocol and its parent Vienna Convention became the UN's first **treaties** to win **universal ratification**.

In January 2023 a UN-backed **assessment forecast recovery** to 1980 levels around 2066 over **Antarctica**, 2045 over the **Arctic**, and 2040 elsewhere. **Healing** will take decades. The sky's **rescue** began with **diligent** observers who trusted that a **baffling** number was no **glitch**, and published it.

中文译文

一层臭氧为地球遮挡了太阳的大部分紫外辐射。1974年6月，马里奥·莫利纳和舍伍德·罗兰，两位化学家，在加州工作，在《自然》杂志上撰文警告，有一种威胁藏在冰箱和喷雾罐里。氯氟烃（CFCs）如此惰性，以至于它们能完好无损地飘入到平流层中，也就是高层大气中，在那里，它们可能释放出氯，氯是一种催化剂，会把臭氧分子一个接一个地破坏掉。两人预料到的，是一场渐进式的侵蚀。

警报真的拉响时，却来得十分突然。1985年5月，约瑟夫·法曼和英国南极调查局的同事报告称，哈雷湾上空的臭氧，那是他们的南极考察站，自20世纪70年代中期起，每到南半球春季就骤降。他们那台不起眼的多布森分光光度计，从地面测量臭氧。工作人员持之以恒，读数时遵循一套一丝不苟的例行规程，全靠人工观测。

然而，美国国家航空航天局（NASA）的雨云7号卫星，这只精密的眼睛在轨道上运行，也记录到了同样的低值。它的软件把这些读数标记为可疑，因为这么低的臭氧此前从没有被可靠地测到过。据NASA的说法，它还在核实这些读数，就在法曼发表论文的时候。传感器并没有出问题；预期却出了问题。

引人注目的是，各国的反应十分迅速。1987年9月，外交官们签署了《蒙特利尔议定书》，这是一项具有约束力的协定，用来限制CFCs。此后的修正案又收紧了它的管控：1990年6月，富裕国家承诺在2000年前逐步淘汰CFCs。2009年9月，该议定书与其所依据的《维也纳公约》一道，成为联合国首批获得普遍批准的条约。

2023年1月，一份由联合国支持的评估报告预测，臭氧层将恢复到1980年的水平：约2066年在南极上空，2045年在北极上空，其他地区则在2040年。臭氧层的愈合还需要数十年。天空得以拯救，始于一群勤勉的观测者，他们相信，一个令人费解的数字并不是小故障，于是把它发表了出来。

词汇与表达

layer n.
层

ozone n.
臭氧
layer of ozone shields Earth from much

shields from v.
为……遮挡了

ultraviolet adj.
紫外

radiation n.
辐射

menace n.
威胁
of a menace in fridges and spray cans.

spray n.
喷雾

inert adj.
惰性
inert that they drift intact into the

intact adj.
完好无损地
inert that they drift intact into the

atmosphere n.
大气

liberate v.
释放
where they can liberate chlorine,

chlorine n.
氯

catalyst n.
催化剂
catalyst that destroys ozone molecule

molecule n.
分子

gradual adj.
渐进式的

erosion n.
侵蚀
The pair foresaw gradual erosion.

alarm n.
警报

abrupt adj.
突然
alarm, when it sounded, was abrupt.

Antarctic adj.
南极

plummeted v.
骤降
plummeted each southern spring since

词汇与表达（续）

humble adj.
不起眼的

gauged v.
测量

persistent adj.
持之以恒
The staff were persistent, reading it

painstaking adj.
一丝不苟的
it in a painstaking routine of manual

routine n.
例行

manual adj.
人工

observation n.
观测

satellite n.
卫星

sophisticated adj.
精密的
a sophisticated eye in orbit,

in orbit n.
在轨道上运行

flagged as phr.
把……标记为

dubious adj.
可疑
Its software flagged them as dubious,

verifying v.
核实
was still verifying them when Farman

sensor n.
传感器

expectation n.
预期

Remarkably adv.
引人注目的是

swift adj.
迅速

diplomats n.
外交官

Protocol n.
议定书

binding adj.
约束力的
a binding accord to curb CFCs.

词汇与表达（续）

accord n.
协定
a binding accord to curb CFCs.

curb v.
限制
a binding accord to curb CFCs.

Amendments n.
修正案

tightened v.
收紧了

affluent adj.
富裕

pledged v.
承诺
June 1990 affluent countries pledged

phase out phr.
逐步淘汰
countries pledged to phase out CFCs by

treaties n.
条约

universal adj.
普遍

ratification n.
批准
to win universal ratification.

assessment n.
评估
2023 a UN-backed assessment forecast

forecast v.
预测

recovery n.
恢复

Antarctica n.
南极

Arctic n.
北极

Healing n.
愈合

rescue n.
拯救

diligent adj.
勤勉的

baffling adj.
令人费解的
a baffling number was no glitch, and

glitch n.
小故障

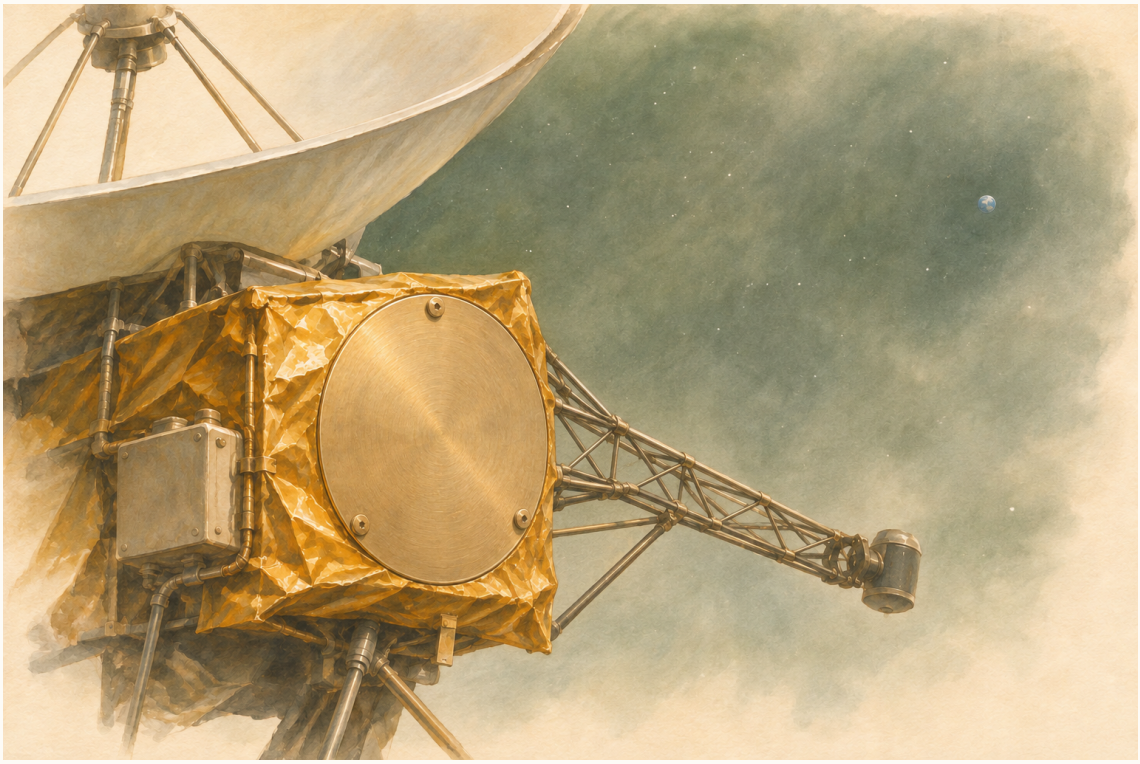
文章简述

- 预警：臭氧层能挡住大量紫外线。1974年，化学家警告：冰箱和喷罐中使用的氯氟烃会飘到高空，释放氯，持续破坏臭氧。
- 发现：1985年，南极地面观测人员报告，哈雷湾臭氧自70年代中期起，每逢南半球春季便骤降。
- 误判：卫星也测到同样低值，软件却因数值太罕见而标为可疑；NASA称当时尚在核实。仪器没有失灵，人的预期才出了错。
- 合作：各国1987年签署《蒙特利尔议定书》，限制氯氟烃；1990年富国承诺2000年前淘汰，议定书2009年获普遍批准。
- 恢复：2023年评估预测，臭氧恢复至1980年水平还需时间：其他地区约2040年、北极约2045年、南极约2066年。

70

写给外星人的一张唱片

The Record Humanity Threw into the Stars



如果有人找到这张唱片，会认识地球吗？

英文原文

In late summer 1977, NASA **fastened** a **disc wrought** from **gilded copper**, 30 **centimetres** wide, to each Voyager **spaceship** before **takeoff**. Nobody could **foresee** who, if anyone, would play one.

Advisers under Carl Sagan of Cornell **distilled** our **earthly abode** into this **compendium**. Voices **greet** the **listener** in 55 languages, followed by the **rumble** of **thunder**, the **whistle** of wind, whales, footsteps, **laughter**, and a **rhythmic heartbeat**. About 90 minutes of **eclectic** music cover **a spectrum of** cultures and **eras**, **notably** Bach and Chuck Berry's Johnny B. Goode.

An **array of** 115 images, **encoded** in analogue form, sits beside printed words from two **statesmen**, Jimmy Carter and Kurt Waldheim. The **sleeve** gives **guidance** on playing the disc, plus a **diagram** fixing the Sun with pulsars as **beacons**.

On 14 February 1990, **at** Sagan's **request**, Voyager 1 **pivoted** its camera for a **glance backwards** across about six billion kilometres. Sixty **frames** formed a **panorama** of the solar system, a **mosaic** in which Earth was a **glimmer** about an eighth of a **pixel across**. They reached Earth's **receivers** between March and 1 May, each **signal** taking five and a half hours.

In a 1994 book, Sagan offered that **pale** dot as a **perspective** on our small and **ephemeral** place amid **cosmic vastness**.

Voyager 1 crossed into **interstellar** space in August 2012, the first **human-made** object there, and Voyager 2 followed in 2018. Both work on with a **longevity unmatched** at NASA, but **depletion** of their power should silence them within a few years.

The Voyagers' discs will **glide on outwards** through the **void** as **messengers** with a map leading back to the Sun.

中文译文

1977年夏末，NASA固定了一张唱片，这个圆盘用镀金的铜锻造而成，宽30厘米，装在每艘旅行者号宇宙飞船上，在起飞之前。谁也无法预见，究竟会不会有人、又会是谁来播放其中一张。

康奈尔大学卡尔·萨根手下的顾问们，把我们在尘世的家园，提炼成这部汇编。人声用55种语言向听者致以问候，接着是雷的隆隆声、风的呼啸声、鲸歌、脚步声、笑声和富有节奏的心跳。约90分钟的兼收并蓄的音乐，涵盖了一系列的文化和时代，尤其是巴赫和查克·贝里的《Johnny B. Goode》。

一系列共115幅图像，以模拟方式编码，旁边是印着的文字，出自于两位政治家，吉米·卡特与库尔特·瓦尔德海姆。（唱片的）套子上给出播放的指引，还有一幅示意图，标定太阳的位置，以脉冲星作信标。

1990年2月14日，应萨根的请求，旅行者1号转动相机，向后投去一瞥，跨越约60亿公里。60帧画面拼成太阳系的全景，在这幅拼图中，地球只是一点微光，宽约八分之一个像素。这些画面抵达地球的接收设备，时间在3月至5月1日间，信号单程要走五个半小时。

在1994年的一本书中，萨根把那个暗淡的小点作为一种视角，借以审视我们渺小而短暂的位置，处在宇宙的浩瀚之中。

旅行者1号进入星际空间，时间是2012年8月，成为第一个到达那里的人造物体；旅行者2号于2018年跟进。两者仍在工作，其长寿在NASA无可匹敌，但电力耗竭预计会让它们在几年之内沉寂。

两艘旅行者号的唱片将继续滑行，向外穿过虚空，作为信使，携着一幅地图，上面的坐标指向太阳。

词汇与表达

fastened v.
固定了
NASA fastened a disc

disc n.
唱片

wrought adj.
锻造
a disc wrought from gilded copper

gilded adj.
镀金的
gilded copper

copper n.
铜

centimetres n.
厘米

spaceship n.
宇宙飞船

takeoff n.
起飞

foresee v.
预见
Nobody could foresee

Advisers n.
顾问

distilled v.
提炼
distilled our earthly abode

earthly adj.
尘世的

abode n.
家园
our earthly abode

compendium n.
汇编
into this compendium

greet v.
向……问候

listener n.
听者

rumble n.
隆隆声

thunder n.
雷

whistle n.
呼啸声

词汇与表达（续）

laughter n.
笑声

rhythmic adj.
富有节奏的

heartbeat n.
心跳

a spectrum of phr.
一系列的

eclectic adj.
兼收并蓄的
About 90 minutes of eclectic music

eras n.
时代

notably adv.
尤其

array of n.
系列

encoded v.
编码
encoded in analogue form

statesmen n.
政治家

sleeve n.
（唱片的）套子

guidance n.
指引
guidance on playing the disc

diagram n.
示意图

beacons n.
信标
with pulsars as beacons

at request phr.
应……请求

pivoted v.
转动
Voyager 1 pivoted its camera

glance n.
一瞥

backwards adv.
向后

frames n.
画面

panorama n.
全景
a panorama of the solar system

词汇与表达（续）

mosaic n.
拼图
a mosaic in which Earth was a glimmer

glimmer n.
微光

pixel n.
像素
an eighth of a pixel across

across adv.
宽

receivers n.
接收设备

signal n.
信号
signal taking five and a half hours

pale adj.
暗淡的

perspective n.
视角
as a perspective

ephemeral adj.
短暂的
our small and ephemeral place

cosmic adj.
宇宙的

vastness n.
浩瀚

interstellar adj.
星际
crossed into interstellar space

human-made adj.
人造

longevity n.
长寿
a longevity unmatched at NASA

unmatched adj.
无可匹敌
a longevity unmatched at NASA

depletion n.
耗竭
depletion of their power

glide on v.
继续滑行

outwards adv.
向外

void n.
虚空

messengers n.
信使

文章简述

- 出发：1977年，两艘旅行者号各带一张镀金铜唱片飞向太空。萨根团队想借它介绍地球，谁会听到、会不会被播放，都无人知道。
- 内容：唱片收录55种语言的问候、自然与生活声音、约90分钟音乐和115幅图像，还附美国总统与联合国秘书长的文字致辞。
- 指路：封套写明怎样播放唱片，并用脉冲星的位置标出太阳在哪里，给潜在听者留下一幅寻找地球所在太阳系的位置图。
- 回望：1990年，旅行者1号从约60亿公里外回拍地球，地球在照片中小得几乎只是一个亮点。萨根借此提醒人类自身的渺小与短暂。
- 远行：1号于2012年进入星际空间，2号2018年跟进。即便它们将来耗尽电力、停止传讯，唱片仍会随探测器继续向外飞行。